

becoming smaller and multilayer to support connectivity with several components at a time. Complexity of the circuitry is increasing and the technology is getting sophisticated. The average thickness of the PCB ranges between 0.5mm and 0.7mm, which, technology experts believe, will decrease further to 0.4mm or less. This will open up new possibilities in the wearables domain.

Traditionally, PCB prepregs have been fibreglass or epoxy, but now substrates are available in several other materials too, such as resin-coated copper, resin-coated film and liquid-crystal film. These prepregs maintain the signal integrity of a circuit and high-end processing at low costs.

Thus, improved PCB substrate and reduced PCB thickness ensure high signal integrity and low output losses, resulting in a reliable and rugged hardware.

Smaller, stronger, faster components. Silicon carbide is becoming the industry-leading technology in power conversion, contributing to the eco-friendly environment. ‘Trench-structure’ design contributes to the long-term reliability of components. Nowadays, these trench-structure MOS devices are acting as alternatives to relays in automobiles (especially electric vehicles) due to their high current handling capability. Their other field applications include industrial automation equipment, solar power inverters, digital power supplies of large-scale servers in data centres and home appliances such as air-conditioners.

Ruggedised power inverters using SiC and IGBT in Power Applications

Niranjan illustrates the example of rooftop solar systems for homes. Anyone can install a roof-top solar panel and generate energy to power different applications. Users can sell off the excess electricity generated to the grid and make money. Power inverters using trench SiC MOSFETs make the power system smart. Higher frequencies result in size reduction of components including magnetics. Reduced size of magnetic components makes the product reliable, efficient and compact. It also lowers the product’s bill-of-materials.

In comparison to conventional FET devices, SiC devices help to cut power losses by 2 per cent, establishing themselves firmly in the market.

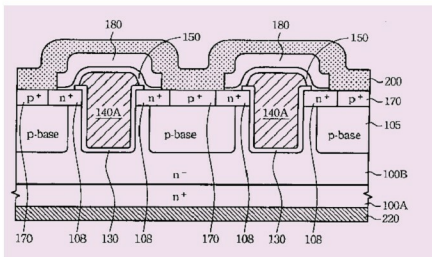


Fig. 2: Trench power MOS in SiC device (Image courtesy: patentimages.storage.googleapis.com)

For low-power and medium-power applications, trench-structure devices provide many advantages:

1. Structural change lowers the drain-to-source resistance (RDS ON) and thus heat dissipation, making the end product efficient
2. Breakdown voltage and switching frequencies of trench-structure MOSFETs and other components using SiC are generally higher than conventional FETs
3. Low on-resistance, high breakdown voltage, high current and high frequencies enable a compact and efficient cooling system
4. The compact size of the component helps in reducing the size and the cost of the PCB

“Thus, silicon carbide reduces the overall system costs and helps in energy conservation by making efficient end-products,” says Niranjan.

Chip resistor—no more laser cuts for accurate value. Niranjan

says, “Thermo-mechanical stress causes resistor failure, so manufacturers are coming up with a technology that allows solder application on the lateral side of the resistor chip.”

Chip resistors’ solder area on their lateral side assists them to sit well on copper pads, avoiding short-circuit and open-circuit faults and breakage due to thermal and mechanical stresses. This helps the resistors to remain cracks-free for a long term.

As far as the manufacturing of chip resistors is concerned, earlier laser cuts were carried out to adjust the resistor value. With that method, usually, the resistor value did not exactly match the customer requirement. Now the practice of laser cuts in resistors has been discontinued, which allows resistor values according to specifications.

Thus, current flows much easily through a component without traces of burns. Change in the structure has increased the reliability and ruggedness of the chip resistor.

Increase in the size of the SiC substrate. The size of the SiC substrate has increased from 5cm to 15cm. The increased size (15cm) is being used in countries like Germany since 2014. It helps in churning more number of different SiC